

Principle part $x-1$. order 1.

$$[674] (a). f(x) = ax + b.$$

$\forall \varepsilon > 0$. Let $\delta = \frac{\varepsilon}{a}$, For any $0 < |x - x_0| < \delta$.

$$\text{s.t. } |f(x) - f(x_0)| = |ax - ax_0| < a \cdot \frac{\varepsilon}{a} = \varepsilon$$

Since $x_0 \in \mathbb{R}$ and x_0 is arbitrary.

$f(x) = ax + b$ is continuous over $\mathbb{R}$.

Date.

(b) $f(x) = x^2$.

$\forall \varepsilon > 0$. Let $\delta = \sqrt{x_0^2 + \varepsilon} - |x_0|$.

For any x with $|x - x_0| < \delta$, we have

$$|f(x) - f(x_0)| = |x^2 - x_0^2| = |x - x_0| \cdot |x + x_0| \leq |x - x_0| (|x - x_0| + 2|x_0|)$$

$$= |x - x_0|^2 + 2|x_0| \cdot |x - x_0|$$

$$< (\sqrt{x_0^2 + \varepsilon} - |x_0|)^2 + 2|x_0| \cdot (\sqrt{x_0^2 + \varepsilon} - |x_0|)$$

$$= \varepsilon.$$

Since x_0 is arbitrary, $f(x)$ is continuous over $\mathbb{R}$.

(c) $f(x) = x^3$.

$\forall \varepsilon > 0$. We want to show $|f(x) - f(x_0)| < \varepsilon$

$$\Leftrightarrow |x^3 - x_0^3| < \varepsilon$$

$$\Leftrightarrow |x - x_0| |x^2 + x_0x + x_0^2| < \varepsilon \Leftrightarrow |x - x_0| (|x|^2 + |x_0||x| + |x_0|^2) < \varepsilon$$

w.d.g. let $|x - x_0| < 1$, then $|x| < 1 + |x_0|$

$$\Leftrightarrow |x - x_0| (|x|^2 + |x_0||x| + |x_0|^2) < \varepsilon$$

$$\Leftrightarrow |x - x_0| < \frac{\varepsilon}{|x|^2 + |x_0||x| + |x_0|^2}$$

$$\text{Let } \delta = \min \left\{ 1, \frac{\varepsilon}{|x|^2 + |x_0||x| + |x_0|^2} \right\}$$

for $0 < |x - x_0| < \delta$, we have $|f(x) - f(x_0)| < \varepsilon$. $\square$

~~For~~

If $x_0 = 0$. Let $\delta = \sqrt{\varepsilon}$.

If $x_0 \neq 0$. we assume $|x - x_0| < \frac{|x_0|}{2}$

$$|x| \leq \frac{3}{2}|x_0|, \text{ i.e. } |x + x_0| \leq \frac{5}{2}|x_0|$$

$$\text{Let } \delta = \min \left\{ \sqrt{\varepsilon}, \frac{|x_0|}{2}, \frac{2\varepsilon}{5|x_0|} \right\}$$

(d) $f(x) = \sqrt{x}$ ($x \geq 0$).

$\forall \varepsilon > 0$, we want to show $|f(x) - f(x_0)| < \varepsilon$.

$$\Leftrightarrow \left| \frac{|x - x_0|}{\sqrt{x} + \sqrt{x_0}} \right| < \varepsilon \Leftrightarrow \frac{|x - x_0|}{\sqrt{x_0}} < \varepsilon$$

1° if $x_0 = 0$, $|\sqrt{x}| < \varepsilon$.

Let $\delta = \varepsilon^2$, for any $0 < |\sqrt{x} - 0| < \delta$, we can $|\sqrt{x} - 0| < \varepsilon$.

2° if $x_0 \neq 0$. Let $\delta = \varepsilon \cdot \sqrt{x_0}$, for any $0 < |\sqrt{x} - \sqrt{x_0}| < \varepsilon$

$$|\sqrt{x} - \sqrt{x_0}| < \frac{|x - x_0|}{\sqrt{x} + \sqrt{x_0}} < \frac{|x - x_0|}{\sqrt{x_0}} < \varepsilon$$

(e) $f(x) = \sqrt[3]{x}$

$\forall \varepsilon > 0$, we want to show $|\sqrt[3]{x} - \sqrt[3]{x_0}| < \varepsilon$

$$\Leftrightarrow \frac{|x - x_0|}{|x^{\frac{2}{3}} + x^{\frac{1}{3}}x_0^{\frac{1}{3}} + x_0^{\frac{2}{3}}|} < \varepsilon \Leftrightarrow \frac{|x - x_0|}{|x_0^{\frac{2}{3}}|} < \varepsilon$$

if $x_0 \neq 0$, Let $\delta = \min(|x_0|^{\frac{1}{3}}, \varepsilon |x_0|^{\frac{2}{3}})$ for any $0 < |x - x_0| < \delta$.

We have $|\sqrt[3]{x} - \sqrt[3]{x_0}| < \varepsilon$ if $|x - x_0| < |x_0|$, $x, x_0 \in \mathbb{R}$.

if $x_0 = 0$, Let $\delta = \varepsilon^3$, for any $0 < |x| < \delta$.

We have $|\sqrt[3]{x}| < \varepsilon$.

Since x_0 is arbitrary and $x_0 \in \mathbb{R}$, $f(x)$ is continuous over $\mathbb{R}$.

(f) $\forall \varepsilon > 0$, we want show $|\sinh x - \sinh x_0| < \varepsilon$.

$$\Leftrightarrow \left| 2 \cos \frac{x_0 + x}{2} \sinh \frac{x - x_0}{2} \right| < \varepsilon$$

$$\Leftrightarrow \left| 2 \sinh \frac{x - x_0}{2} \right| < \varepsilon$$

$$\Leftrightarrow \left| 2 \cdot \frac{x - x_0}{2} \right| < \varepsilon$$

Let $\delta = \varepsilon$. For any $0 < |x - x_0| < \delta$.

We can $|\sinh x - \sinh x_0| < \varepsilon$.

(g) $\forall \varepsilon > 0$. We want $|\cos x - \cos x_0| < \varepsilon$

$$\Leftrightarrow 2 \left| \sin \frac{x+x_0}{2} \right| \cdot \left| \sin \frac{x-x_0}{2} \right| < \varepsilon.$$

$$\Leftrightarrow 2 \left| \frac{\sin x - \sin x_0}{2} \right| < \varepsilon.$$

Let $\delta = \varepsilon$. Similar as (f).

(8) $\forall \varepsilon > 0$. We want to show $|\arctan x - \arctan x_0| < \varepsilon$.

Let $\arctan x = t$ $\arctan x_0 = t_0$.

We want $|t - t_0| < \varepsilon$.

$$\Leftrightarrow |\tan |t - t_0|| < \varepsilon \quad (\text{when } 0 < |t - t_0| < \frac{\pi}{2}).$$

$$\Leftrightarrow \frac{2 |\tan \frac{t+t_0}{2} - \tan \frac{t-t_0}{2}|}{1 - |\tan \frac{t+t_0}{2}| |\tan \frac{t-t_0}{2}|} < \varepsilon$$

$$\Leftrightarrow |\tan t - \tan t_0| < \varepsilon$$

$$\Leftrightarrow |x - x_0| < \varepsilon.$$

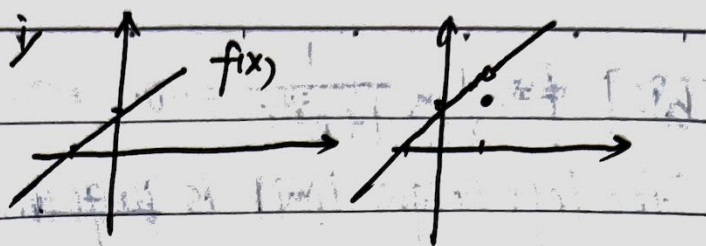
Let $\delta = \min \left\{ \varepsilon, \frac{\pi}{2} \right\}$. When $|x - x_0| < \delta$

We have $|\arctan x - \arctan x_0| < \varepsilon$.

$\therefore \arctan x$ is continuous over $\mathbb{R}$.

$$* |\arctan x - \arctan x_0| = \left| \arctan \frac{x - x_0}{1 + x x_0} \right|$$

[b76]. $f(x) = \begin{cases} x+2 & x \neq 2 \\ A & x = 2 \end{cases}$

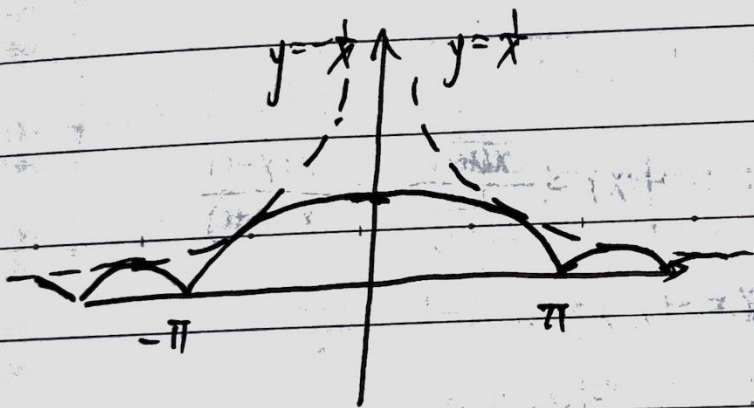


i) $A=2$. continuous over $\mathbb{R}$.

ii) $A \neq 2$. have removable discontinuity point $x=2$.
continuous over $(-\infty, 2) \cup (2, +\infty)$.

[b78] (i). $\lim_{x \rightarrow 0} \frac{\sin x}{x} = f_1(0) = 1$. $f_1(x)$ is continuous in $x=0$.

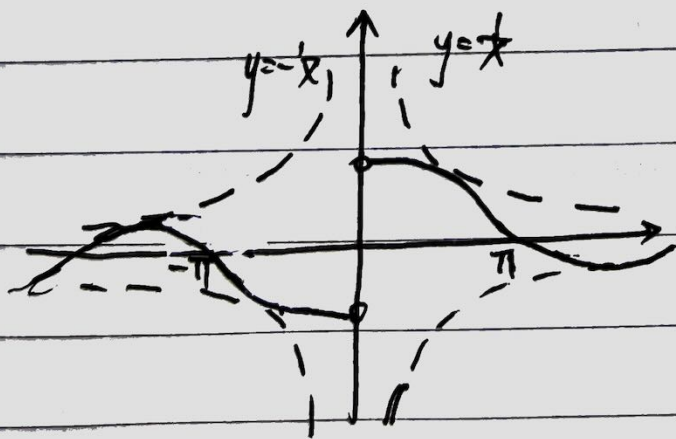
$f_1(x) = \frac{\sin x}{x}$ ($x \neq 0$) is continuous. thus. $f_1(x)$ is continuous over $\mathbb{R}$.



(2) $\lim_{x \rightarrow 0^+} \frac{\sin x}{|x|} = 1$ $\lim_{x \rightarrow 0^-} \frac{\sin x}{|x|} = -1$

the $x=0$ is a jump discontinuity of $f_2(x)$.

$f_2(x)$ is continuous over $(-\infty, 0) \cup (0, +\infty)$

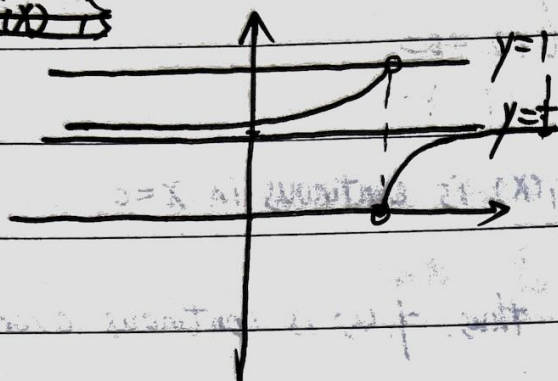


[682] ~~f(x)~~ $\lim_{x \rightarrow 1^+} \frac{1}{1+e^{x-1}} = 0$ $\lim_{x \rightarrow 1^-} \frac{1}{1+e^{x-1}} = 1$

the left/right limit is ~~unequal~~ unequal.

$f(x)$ is continuous over $(-\infty, 1) \cup (1, +\infty)$

~~f(x)~~



$$\lim_{x \rightarrow -\infty} f(x) = \frac{1}{2}$$

$$\lim_{x \rightarrow +\infty} f(x) = \frac{1}{2}$$

[690] $y = \frac{\frac{1}{x} - \frac{1}{x+1}}{\frac{1}{x+1} - \frac{1}{x}} = f(x)$ $f(x) = \frac{\frac{x(x+1)}{x(x+1)}}{\frac{x(x+1)}{x(x+1)}} = \frac{x(x-1)}{x(x+1)}$

we need to consider $x = -1$, $x = 0$, $x = 1$.

1. $x = 1$ $\lim_{x \rightarrow 1^+} f(x) = 0$ $\lim_{x \rightarrow 1^-} f(x) = 0$

removable discontinuity point

2. $x = 0$ $\lim_{x \rightarrow 0^-} f(x) = -1$ $\lim_{x \rightarrow 0^+} f(x) = -1$

3. $x = -1$ $x \rightarrow -1^-$, $f(x) \rightarrow +\infty$

$x = 1$, $x = 0$. First kind removable discontinuity point

$x = -1$. Second kind

[698] $y = e^{x+\frac{1}{x}}$ we shall consider $x \rightarrow 0$

$\lim_{x \rightarrow 0^+} e^{x+\frac{1}{x}} = +\infty$ doesn't exist. second kind.

[740]. We should calculate the limit at $x \rightarrow 0$.

$$(1) \lim_{x \rightarrow 0} \frac{\frac{1}{2}x}{\frac{1}{3}x} = \frac{3}{2}$$

$$(2) \lim_{x \rightarrow 0} f(x) = \frac{2x}{x} = 2.$$

$$(3) \lim_{x \rightarrow 0} f(x) = x \cdot \sin \frac{1}{x} = 0.$$

$$(4) \lim_{x \rightarrow 0} f(x) = e.$$

$$(5) \lim_{x \rightarrow 0} f(x) = \frac{1}{e^{\frac{1}{x}} \cdot x^2} \xrightarrow{t=\frac{1}{x}} \lim_{t \rightarrow \infty} \frac{t}{e^t} = 0.$$

$$(6) \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} e^{x \ln x} = \lim_{x \rightarrow 0} e^{\frac{-\ln x}{\frac{1}{x}}} = e^{\lim_{x \rightarrow 0} \frac{-\ln x}{\frac{1}{x}}} = 1$$

$$(7) \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{-2 \ln x}{\frac{1}{x}} \xrightarrow{t=\frac{1}{x}} \lim_{t \rightarrow \infty} \frac{-2 \ln t}{t} = 0.$$

[148]. $f(x)$ is continuous ^{over} in $[a, b]$.

Verify. $m(x) = \inf_{a \leq \xi \leq x} \{f(\xi)\}$ and $M(x) = \sup_{a \leq \xi \leq x} \{f(\xi)\}$ is continuous over $[a, b]$.

Proof: $\forall \epsilon > 0, \exists \delta > 0, 0 < |x - x_0| < \delta$.
 $\forall x_0 \in [a, b]$
 1° first we show the ~~left~~ ^{right} continuity.

$$x_0 < x < x_0 + \delta \Rightarrow |f(x_0) - f(x)| < \epsilon$$

$$\text{i.e. } (f(x) < \epsilon + f(x_0)) \wedge (f(x) > f(x_0) - \epsilon)$$

When $a \leq x \leq x_0$, $f(x) \geq m(x) > m(x_0) - \epsilon$. ^{$f(x)$ is the least value}

Thus, we have $m(x) \geq m(x_0) - \epsilon$. ($x_0 < x < x_0 + \delta$)

Obviously, $m(x)$ is nonincreasing $\epsilon + m(x_0) \geq m(x) \geq m(x_0) - \epsilon$.

$$\Rightarrow |m(x_0) - m(x)| < \epsilon.$$

2° then show the left continuity.

w.l.g. we assume $m(x_0) = f(x_0)$ when $x \in [a, x_0]$

(otherwise, $m(x_0) = f(x_0)$, $a \leq x_1 < x_0$, then for $x_1 < x < x_0$, $m(x) = m(x_0)$
 x is left continuous.)

$$\forall \epsilon > 0, \exists \delta > 0, x_0 - \delta < x < x_0.$$

We have $f(x) < f(x_0) + \epsilon = m(x_0) + \epsilon$.

$$\Leftrightarrow m(x_0) \leq m(x) < m(x_0) + \epsilon \Leftrightarrow |m(x) - m(x_0)| < \epsilon.$$

proof 2: $\forall x_0 \in [a, b]$

1° first we show the left continuity. w.l.g. we assume $M(x_0) = f(x_0)$

$$\forall \epsilon > 0, \exists \delta > 0, x_0 - \delta < x < x_0. \text{ we have } f(x_0) - \epsilon < f(x) < f(x_0) + \epsilon$$

$M(x) < M(x_0) + \epsilon$. We need to show $M(x) > M(x_0) - \epsilon$

$$\text{When } x \in [a, x_0], f(x) > f(x_0) - \epsilon = M(x_0) - \epsilon.$$

$$\therefore M(x_0) - \epsilon < M(x) < M(x_0) + \epsilon.$$

2° then show the right continuity.

~~Similarly as proof 1. w.t.g. we assume $M(x_0) = f(x_0)$ ($x \in [a, x_0]$).~~

$$\forall \varepsilon > 0, \exists \delta > 0, \quad x_0 - \delta < x < x_0 + \delta, \quad f(x_0) - \varepsilon < f(x) < f(x_0) + \varepsilon.$$

$$\text{We have } f(x) > f(x_0) > f(x_0) - \varepsilon = M(x_0) - \varepsilon. \quad M(x) > M(x_0) - \varepsilon$$

$$\text{when } x \in [a, x_0], \quad f(x) \leq M(x) < M(x_0) + \varepsilon$$

$$> x \in (x_0, x_0 + \delta), \quad f(x) < f(x_0) + \varepsilon < M(x_0) + \varepsilon.$$

$$\text{Hence, } M(x_0) + \varepsilon > f(x) > M(x_0) - \varepsilon.$$

[750] Proof. By the definition of ~~sup~~ $M(x)$,

$$\forall x_0 \in (a, b).$$

$$\forall \varepsilon > 0, \text{ ~~there exists~~ there } \exists t_0 \in (a, x_0)$$

$$f(t_0) > M(x_0) - \varepsilon.$$

$$\text{Let } \delta = x_0 - t_0, \text{ when } x_0 - \delta = t_0 < x < x_0.$$

$$M(x_0) - \varepsilon < f(t_0) \leq M(x) = \sup_{a \leq t < x} \{f(t)\} \leq M(x_0)$$

$$\therefore \lim_{x \rightarrow x_0^-} M(x) = M(x_0)$$

Similarly, there $\exists t_1 \in (a, x_0)$.

$$f(t_1) < m(x_0) + \varepsilon.$$

$$\text{Let } \delta = x_0 - t_1, \text{ when } x_0 - \delta = t_1 < x < x_0.$$

$$m(x_0) + \varepsilon > f(t_1) \geq m(t_1) \geq m(x) \geq m(x_0)$$

$$\therefore \lim_{x \rightarrow x_0^-} m(x) = m(x_0).$$

[751] Let $\lim_{x \rightarrow \infty} f(x) = A$. $|f(x) - A| < \epsilon$ 可以不合上下界

$\forall \epsilon > 0, \exists X > 0$ s.t. $|f(x) - A| < \epsilon$ for any $x > X$.

$|f(x) - A| < \epsilon \Leftrightarrow$ i.e. $f(x) < A + \epsilon$ $f(x) > A - \epsilon$.

Now consider the closed interval $[a, X]$

Since the $f(x)$ is continuous on $[a, X]$, we must have maximum and minimum, let them be M and m .

Let $M' = \max \{A + \epsilon, M\}$ for any $x \in [a, +\infty)$ s.t. $f(x) \leq M'$

$m' = \min \{A - \epsilon, m\}$, for any $x \in [a, +\infty)$ s.t. $f(x) \geq m'$

M' is a upper bound, m' is a lower bound. thus...

[788] $f(x) = \frac{1}{x}$

proof: continuity: for any $x_0 \in (0, 1)$

$\forall \epsilon > 0$ we want to show $|f(x) - f(x_0)| < \epsilon$ for any $|x - x_0| < \delta$.

$$\Leftrightarrow \frac{|x - x_0|}{|x_0 x|} < \epsilon$$

Since $y = x$ is continuous on $(0, 1)$ and $y \neq 0$.

$f(x) = \frac{1}{x}$ is defined in $(0, 1)$, thus the $f(x) = \frac{1}{x}$ is continuous on $(0, 1)$

not uniformly continuous:

we consider a pair of points $x_{n1} = \frac{1}{n}$ $x_{n2} = \frac{1}{n+1}$.

Since $n \rightarrow +\infty$, $|x_{n1} - x_{n2}| \rightarrow 0$.

However $|f(x_{n1}) - f(x_{n2})| = 1$.

Let $\epsilon_0 = \frac{1}{2}$, for any $\delta > 0$, we can find $|f(x_{n1}) - f(x_{n2})| \geq \epsilon_0$

s.t. $|x_{n1} - x_{n2}| < \delta$, thus...

[789] Proof: Since $y_1 = \sin x$ is continuous on $(0, 1)$, $y_2 = \frac{\pi}{x}$ is continuous.

$f(x)$ can be considered as the composite of y_1 and y_2 .

$f(x)$ is continuous over $(0, 1)$

2° $x \in (0, 1) \Rightarrow \frac{\pi}{x} \in (\pi, +\infty)$.

For any $x \in (0, 1)$, $\sin(\frac{\pi}{x}) \in (-1, 1)$. Thus, bounded.

3° We can find a pair of point $x_{n_1} = \frac{1}{2n}$, $x_{n_2} = \frac{1}{n + \frac{1}{2}}$

since $n \rightarrow +\infty$, $|x_{n_1} - x_{n_2}| \rightarrow 0$.

However, for any $n \in \mathbb{N}$, we have $|f(x_{n_1}) - f(x_{n_2})| \equiv 1$

similar as [788], the $f(x) = \sin \frac{\pi}{x}$ is not uniformly continuous on $(0, 1)$

[791] Proof:

① Let $\lim_{x \rightarrow +\infty} f(x) = A$. $\forall \varepsilon > 0$, $\exists X > 0$, s.t. $|f(x) - A| < \frac{\varepsilon}{2}$ for any $x > X$.

② By the Cantor-Heine's theorem, $f(x)$ is uniformly continuous on $[a, X]$. i.e. there exist $\delta_1 > 0$, s.t. $|f(x_1) - f(x_2)| < \varepsilon$ for any $a \leq x_1 - x_2 < \delta_1$.
 $\downarrow$ $[a, X+1]$ $\forall x \in [a, X+1]$ $\forall \delta(x)$

③ There exist $\delta_2 > 0$ for any x_1, x_2 such that: $(x_1 > X \wedge x_2 > X)$, and $0 < |x_1 - x_2| < \delta_2$, such that $|f(x_1) - f(x_2)| \leq |f(x_1) - A| + |f(x_2) - A| < \varepsilon$.

④ By the local properties of continuity, consider the point X .
 ~~There~~ there exists $\delta_3 > 0$, s.t. $\omega(f|_{U_{\delta_3}(X)}) < \varepsilon$.

i.e. for any $x_1, x_2 \in U_{\delta_3}(X)$, $|f(x_1) - f(x_2)| < \varepsilon$.

Let $\delta = \min\{\delta_1, \delta_2, \delta_3\}$ for any $x_1, x_2 \in [a, +\infty)$ w.l.g. let $x_1 < x_2$.

x_1, x_2 subject to one of these cases: ① $x_1, x_2 \in [a, X]$, ② $x_1, x_2 \in [X, +\infty)$

③ $x_1 \in [a, X] \wedge (x_2 \in [X, +\infty))$, which all s.t. $|f(x_1) - f(x_2)| < \varepsilon$ when $|x_1 - x_2| < \delta$

[792]. Proof: ~~For any~~

$\forall \varepsilon > 0$. $\exists \delta = \frac{\varepsilon}{2}$ for any $x_1, x_2 \in \mathbb{R}$ s.t. $|x_1 - x_2| < \delta$.

We ~~can~~ have $|f(x_1) - f(x_2)| = |(x_1 - x_2) + (\sin x_1 - \sin x_2)|$
 $\leq |x_1 - x_2| + |2 \sin \frac{x_1 - x_2}{2} \cos \frac{x_1 + x_2}{2}| \leq 2|x_1 - x_2| < 2\delta = \varepsilon$.

Thus, the function $f(x) = x + \sin x$ is uniformly continuous over $\mathbb{R}$.

Nov. 3rd.

ob. proof: Let $\lim_{x \rightarrow \infty} f(x) = A$.

Let $\varepsilon_0 = 1$. $\exists X \in \mathbb{R} [a, +\infty)$ s.t. $|f(x) - A| < \varepsilon_0$, for any $x > X$.

i.e. $|f(x)| < |A| + 1$

Now consider the closed interval $[a, X+1]$, By the Weierstrass maximum-value theorem, there exists maximum and minimum values M and m of $f(x)$ on $[a, X+1]$

$M' = \max\{M, |A| + 1\}$, $f(x) < M'$ (for $x \in [a, +\infty)$)

$m' = \min(-|A| - 1, m)$, $f(x) > m'$ (for $x \in [a, +\infty)$)

$\therefore f(x)$ is bounded on $[a, +\infty)$

1° If there exist $x_0 \in [a, +\infty)$ s.t. $f(x_0) > A$. We can proof $f(x)$ has maximum on $[a, +\infty)$.
 实质是讨论极限与确界的联系 (有最大值一定是确界)

Let $\varepsilon_0 = \frac{f(x_0) - A}{2}$. $\exists X \in [a, +\infty)$ s.t. $|f(x) - A| < \varepsilon_0$ for $x > X$

i.e. $\frac{3A - f(x_0)}{2} < f(x) < \frac{f(x_0) + A}{2} < f(x_0)$

we have $A < f(x) < f(x_0)$

Now consider the closed interval $[a, x_1]$ must have a maximum M .
Further, $M \geq f(x_0) > f(x)$ (for $x > x_1$). M is a maximum point of $f(x)$. $x \in [a, \infty)$

1° If for any $x \in [a, +\infty)$ s.t. $f(x) \leq A$, then we have a minimum point m of $f(x)$.

~~∵ $\lim_{x \rightarrow +\infty} f(x) = A$. Let $\varepsilon_0 = 1$. $\exists x_2 \in [a, +\infty)$ s.t. $|f(x) - A| < \varepsilon_0$ for $x > x_2$~~
i.e. ~~$A - 1 < f(x) < A$~~

2° If $\exists x_0 \in [a, +\infty)$ s.t. $f(x_0) < A$, ~~we have~~ $f(x)$ has minimum point.

Let $\varepsilon_0 = \frac{A - f(x_0)}{2}$ $\exists x_2 \in [a, +\infty)$ s.t. $|f(x) - A| < \varepsilon_0$ for $x \in x_2$.

$$\text{i.e. } \frac{f(x_0) + A}{2} < f(x) < \frac{3A - f(x_0)}{2}$$

$$\text{i.e. } f(x_0) < f(x) < A$$

similar as 1. we can find a minimum point m for the $f(x)$, $x \in [a, +\infty)$

o 12. $f(x) = \sqrt{x}$.

1° first we consider the close interval $[0, 2]$.

the elementary fundamental function is continuous on $[0, 2]$.

By Cantor-Heine theorem. it's uniformly continuous.

2° now we consider the ~~close interval~~ ^{open interval} $[1, +\infty)$

For any $\varepsilon > 0$, Let $\delta_i = \varepsilon$, we have

$$|f(x_1) - f(x_2)| = \left| \frac{x_1 - x_2}{\sqrt{x_1} + \sqrt{x_2}} \right| < |x_1 - x_2| < \varepsilon \quad (\text{for any } x_1, x_2)$$

such that $|x_1 - x_2| < \delta$.

Let $\delta = \min \{ \delta_i, 1 \}$ for any $x_1, x_2 \in [0, +\infty)$ s.t. $|x_1 - x_2| < \delta$.

We have $|f(x_1) - f(x_2)| < \varepsilon$.

Date.

014 proof: ~~$\forall \varepsilon > 0$~~ . Let $\delta = \frac{\varepsilon}{L}$. For any $|x' - x''| < \delta$.
we have $|f(x') - f(x'')| \leq |x' - x''| \cdot L < L \cdot \frac{\varepsilon}{L} = \varepsilon$. ($L > 0$)
Hence, $f(x)$ is uniformly continuous

Ex 015. $f(x) = \sin x$.

Proof: $\forall \varepsilon > 0$. Let $\delta = \varepsilon$. For any $|x_1 - x_2| < \delta$ ($x_1, x_2 \in \mathbb{R}$)

$$|\sin x_1 - \sin x_2| = 2 \left| \sin \frac{x_1 - x_2}{2} \cos \frac{x_1 + x_2}{2} \right| \leq 2 \sin \frac{|x_1 - x_2|}{2} \leq |x_1 - x_2| < \delta = \varepsilon$$

i.e. $f(x)$ is uniformly continuous over $\mathbb{R}$.